

DAY 07

Windows Failed Logon Investigation

Professional SOC L1 Investigation Report

Field	Value
Case ID	AUTH-2026-001
Detection verdict	True Positive
Activity classification	Benign authentication failure
Severity / status	Low / Closed after remediation

Safety and scope	This is a synthetic, sanitized training case. Hostnames, accounts, timestamps, and application names are fictional. No live system was accessed.
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Executive Summary

A detection rule identified 18 failed Windows network logons from an internal application server to a workstation over a three-minute period. Investigation of authentication, endpoint, and change-management evidence determined that an approved backup service was repeatedly using stale stored credentials after the account password had changed. No successful unauthorized authentication, malware, or suspicious network activity was identified. Updating the stored credentials stopped the failures and restored the backup job.

Analyst conclusion	The alert was a true positive because the repeated failures occurred, but the underlying activity was benign and non-malicious.
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Case Overview

Field	Value
Alert	Multiple Failed Windows Logins
Target account	jrahman
Source	NS-APP-02 - internal application server
Destination	NS-WKS-104
Authentication	Event ID 4625, Logon Type 3
Volume	18 failures in 3 minutes
Network service	SMB over TCP port 445

Investigation Workflow

Step	Evidence reviewed	Finding
1. Validate alert	Event ID, account, count, time window	Repeated authentication failures were confirmed.
2. Identify logon context	Logon Type 3	Attempts were network authentications, not local console or RDP logons.
3. Enrich source	Asset inventory and network zone	Source mapped to an internal application server owned by IT Operations.
4. Correlate success	Event ID 4624 and 4740	No matching successful logon or account lockout was observed.
5. Analyze pattern	Failure timestamps	Attempts occurred at regular 10-second intervals, indicating automation.
6. Review endpoint process	EDR process and connection telemetry	Approved, signed BackupAgent.exe initiated SMB connections as a Windows service.
7. Confirm root cause	Identity and change records	The account password changed, but the backup service retained the old password.
8. Validate remediation	Post-change authentication and job status	Credential update stopped failures and the backup completed successfully.

Evidence Summary

Evidence	Result	Assessment
Failed logons	18 x 4625	Real activity; detection was valid
Successful logon	None from same source/account in window	No evidence that the attempts succeeded
Account lockout	None	Threshold was not reached
Process	Approved BackupAgent.exe	Expected software and path
Digital signature	Valid	Supports legitimacy but was not treated as conclusive alone
Retry pattern	Every 10 seconds	Automated service behavior
Threat evidence	None found	No malware, anomalous process, or unauthorized access

Timeline

Time (EDT)	Event
09:05	The account password was changed.
09:14-09:17	BackupAgent.exe generated 18 failed Type 3 logons using the old stored password.
09:27	IT Operations updated the stored credentials.
After 09:27	Authentication failures stopped and the backup job completed successfully.

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Element	Finding
Who	The jrahman account was used by BackupAgent.exe on NS-APP-02 to authenticate to NS-WKS-104.
What	The approved application generated 18 failed Type 3 network logons using stale stored credentials.
When	The failures occurred from 09:14 to 09:17 EDT on September 22, 2026. Remediation occurred at 09:27 EDT.
Where	The activity originated from NS-APP-02 and targeted NS-WKS-104 over SMB on TCP port 445.
Why	The account password changed, but the backup application retained the old password and retried every 10 seconds.

Final Determination

Decision field	Determination
Detection verdict	True Positive
Activity classification	Benign authentication failure
Root cause	Stale credentials stored in an approved backup service
Severity	Low
Account compromise	Not observed
Malicious activity	Not observed
Status	Closed after remediation

Remediation and Recommendations

- Updated the credentials stored by the backup application and confirmed successful job completion.
- Recommended replacing the personal user account with a dedicated service identity.
- Recommended a managed service account or approved credential vault where supported.
- Recommended a runbook for updating dependent services after password changes.
- Recommended monitoring for recurrence of the same source-account authentication pattern.

Key Lessons

- A single Event ID 4625 does not prove brute force activity.
- Logon Type identifies the authentication context and must be interpreted with other evidence.
- Regular retry intervals suggest automation but do not identify whether the automation is benign or malicious.
- A true-positive detection can represent benign activity when the rule correctly identifies the behavior it was designed to detect.
- Closure requires documented root cause, remediation, and confirmation that suspicious activity did not continue.

Portfolio scope

This report demonstrates a guided SOC Level 1 workflow using synthetic and sanitized evidence. It does not claim access to a production environment.